

Packet Tracer - Configure Initial Router Settings

Objectives

Part 1: Verify the Default Router Configuration

Part 2: Configure and Verify the Initial Router Configuration

Part 3: Save the Running Configuration File

Background

In this activity, you will perform basic router configuration tasks. You will secure access to the CLI and console port using encrypted and plain-text passwords. You will also configure messages for users who are logging into the router. These banners warn unauthorized users that access is prohibited. Finally, you will verify and save your running configuration.

Instructions

Part 1: Verify the Default Router Configuration

Step 1: Establish a console connection to R1.

- Choose a **Console** cable from the available connections.
- Click **PCA** and select **RS 232**.
- Click **R1** and select **Console**.
- Click **PCA > Desktop** tab > **Terminal**.
- Click **OK** and press **ENTER**. You are now able to configure **R1**.

Step 2: Enter privileged mode and examine the current configuration.

You can access all the router commands from privileged EXEC mode. However, because many of the privileged commands configure operating parameters, privileged access should be password-protected to prevent unauthorized use.

- Enter privileged EXEC mode by entering the **enable** command.

```
Router> enable
Router#
```

Notice that the prompt changed in the configuration to reflect privileged EXEC mode.

- Enter the **show running-config** command.

```
Router# show running-config
```

What is the router's hostname?

Router

How many Fast Ethernet interfaces does the Router have?

4 Fast Ethernet interfaces

How many Gigabit Ethernet interfaces does the Router have?

2 Gigabit Ethernet interfaces

How many Serial interfaces does the router have?

2 Serial interfaces

What is the range of values shown for the vty lines?

0 through 4

- c. Display the current contents of NVRAM.

```
Router# show startup-config
startup-config is not present
```

Why does the router respond with the **startup-config is not present** message?

We haven't configured it yet.

Part 2: Configure and Verify the Initial Router Configuration

To configure parameters on a router, you may be required to move between various configuration modes. Notice how the prompt changes as you navigate through the IOS configuration modes.

Step 1: Configure the initial settings on R1.

Note: If you have difficulty remembering the commands, refer to the content for this topic. The commands are the same as you configured on a switch.

- Configure **R1** as the hostname.
- Configure Message of the day text: **Unauthorized access is strictly prohibited.**
- Encrypt all plain text passwords.

Use the following passwords:

- Privileged EXEC, unencrypted: **cisco**
- Privileged EXEC, encrypted: **itsasecret**
- Console: **letmein**

Step 2: Verify the initial settings on R1.

- Verify the initial settings by viewing the configuration for R1.

What command do you use?

show running-config

- Exit the current console session until you see the following message:

```
R1 con0 is now available
```

```
Press RETURN to get started.
```

- Press **ENTER**; you should see the following message:

```
Unauthorized access is strictly prohibited.
```

```
User Access Verification
```

```
Password:
```

Why should every router have a message-of-the-day (MOTD) banner?

In order to let the user know about if any legal notices, any issues regarding the system, and other general updates.

If you are not prompted for a password before reaching the user EXEC prompt, what console line command did you forget to configure?

"login"

- d. Enter the passwords necessary to return to privileged EXEC mode.

Why would the **enable secret** password allow access to the privileged EXEC mode and the **enable password** no longer be valid?

The secret password is encrypted, so it has a higher level of security. Additionally, because it is encrypted, you still can't see it in the running-config output.

If you configure any more passwords on the router, are they displayed in the configuration file as plain text or in encrypted form? Explain.

They will be encrypted, "service password-encryption" applies to current and future passwords.

Part 3: Save the Running Configuration File

Step 1: Save the configuration file to NVRAM.

- a. You have configured the initial settings for **R1**. Now back up the running configuration file to NVRAM to ensure that the changes made are not lost if the system is rebooted or loses power.

What command did you enter to save the configuration to NVRAM?

copy running-config startup-config

What is the shortest, unambiguous version of this command?

cop r st

Step 2: Optional: Save the startup configuration file to flash.

Although you will be learning more about managing the flash storage in a router in later chapters, you may be interested to know that, as an added backup procedure, you can save your startup configuration file to flash. By default, the router still loads the startup configuration from NVRAM, but if NVRAM becomes corrupt, you can restore the startup configuration by copying it over from flash.

Complete the following steps to save the startup configuration to flash.

- a. Examine the contents of flash using the **show flash** command:

R1# **show flash**

How many files are currently stored in flash?

3 files

Which of these files would you guess is the IOS image?

Probably c1900-universalk9-mx.SPA.151-4.M4.bin

```
System flash directory:
File      Length  Name/status
  3    33591768  c1900-universalk9-mz.SPA.151-4.M4.bin
  2     28282    sigdef-category.xml
  1     227537    sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

Why do you think this file is the IOS image?

The size is much larger than the other two, and it is a bin file compared to xml

```
R1# copy startup-config flash
```

```
Destination filename [startup-config]
```

The router prompts you to store the file in flash using the name in brackets. If the answer is yes, then press **ENTER**; if not, type an appropriate name and press **ENTER**.

- b. Use the **show flash** command to verify the startup configuration file is now stored in flash.

```
System flash directory:
File  Length  Name/status
 3   33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
 2    28282  sigdef-category.xml
 1    227537  sigdef-default.xml
 4     1264   startup-config
[33848851 bytes used, 221895149 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```